WEEK 9 · LECTURE L1 OF 3 · UNIT III

Algorithmic Decision-Making in HR

Where algorithms now make or shape HR decisions, and what the Amazon case teaches.

Part 1	Screening, matching, appraisal, promotion, attrition prediction
Part 2	The Amazon hiring tool: a cautionary tale
Part 3	Debrief against what pods found in Round 4
Read	Tambe, Cappelli and Yakubovich (2019); Kellogg, Valentine and Christin (2020)

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PART 1 · WHERE ALGORITHMS ACT

Five HR decisions algorithms now shape

Decision	What the algorithm does	The risk
Screening	Ranks or filters applicants	Learns past bias; unknown validity
Matching	Suggests roles, shifts or orders	Feedback loops that lock people in
Appraisal	Scores performance from activity data	Measures what is easy, not what matters
Promotion	Ranks readiness	Proxies for tenure, gender or location
Attrition prediction	Flags who may leave	False alarms that change how people are treated

PART 2 · THE AMAZON CASE

A model that learned the past

What happened

Amazon built an experimental tool to score resumes. It was scrapped after it was found to downgrade resumes mentioning women's activities (Reuters, 2018).

Why

It was trained on ten years of resumes from mostly male hires, so it treated male-linked signals as merit.

Why fixes failed

Removing gender words did not help: other features acted as proxies.

THE LESSON A model trained on past decisions inherits their bias, even if no one intended it.

PART 3 · ROUND 4 DEBRIEF

GreenLeaf's flight-risk model, by age

Measure	Under 30	30 and over
Share flagged High	72.6%	24.6%
False positive rate (stayers wrongly flagged)	70.1%	22.7%
Precision (flags that are right)	25.4%	25.9%
Actual exit rate	22.8%	19.6%

READ IT

Equally precise for both groups, yet it flags young staff three times as often and wrongly flags 70% of young stayers. Source: PeopleAnalytics_Fact, Age, Predicted_Attrition, Actual_Attrition.

Going deeper: the theory underneath algorithmic decisions

The next slides set out Kellogg et al.'s six Rs, Tambe et al.'s four challenges, algorithm aversion, and algorithm appreciation: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · KELLOGG, VALENTINE AND CHRISTIN (2020)

The six Rs of algorithmic control

THE CORE IDEA

Algorithms control work in new ways, through three mechanisms: direction, evaluation and discipline.

HOW IT WORKS

Restricting and recommending (direct); recording and rating (evaluate); replacing and rewarding (discipline).

LIMITS AND CRITIQUE

Workers resist through what the authors call algoactivism, so control is never complete.

AT GREENLEAF Gig allocation scores at GreenLeaf restrict, rate and reward partners at once.

THEORY IN DEPTH · TAMBE, CAPPELLI AND YAKUBOVICH (2019)

Four reasons AI is hard in HR

THE CORE IDEA

HR outcomes are complex, data sets are small, fairness and law constrain models, and employees react to being judged.

HOW IT WORKS

They recommend causal reasoning, experiments, and involving employees in design.

LIMITS AND CRITIQUE

Written before large language models changed what AI can do with text.

AT GREENLEAF 1,000 employees is a small data set. A model trained on it will be unstable for small groups.

THEORY IN DEPTH · DIETVORST, SIMMONS AND MASSEY (2015)

People abandon algorithms after one mistake

THE CORE IDEA

Algorithm aversion: after seeing a model err, people prefer human judgment even when the model is more accurate.

HOW IT WORKS

People forgive human errors more readily than machine errors.

LIMITS AND CRITIQUE

Letting people slightly adjust the model's output reduces aversion.

AT GREENLEAF

Managers who see one bad flight-risk flag may stop trusting all analytics, good and bad alike.

THEORY IN DEPTH · LOGG, MINSON AND MOORE (2019)

People also over-trust algorithms

THE CORE IDEA

Algorithm appreciation: lay people often weight advice more when told it comes from an algorithm.

HOW IT WORKS

Trust is higher for numeric estimates, and lower among experts.

LIMITS AND CRITIQUE

Aversion and appreciation coexist; which wins depends on the task and the person.

AT GREENLEAF

A flight-risk score shown as a precise number may be trusted more than it deserves. Show the error rate beside it.

TAKEAWAYS

Three things to carry out of this lecture

1

Every HR algorithm carries a specific risk; name it.

2

Models trained on past decisions inherit past bias.

3

Precision can be equal while error rates differ sharply.

NEXT Fairness definitions, and why they conflict.